Causal Adversarial Autoencoder for Disentangled SAR Image Representation and Few-Shot Target Recognition

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***Abstract*— Lack of interpretability and weak generalization ability have become the major challenges with data-driven intelligent synthetic aperture radar-automatic target recognition (SAR-ATR) technology, especially in practical applications with azimuth-sparse training samples. A novel insight into SAR image representation with neural networks from a causal perspective is presented in this article. First, a causal model of SAR image representation conditioned on disentangled semantic factors is proposed. A set of SAR images is considered as a low-dimensional manifold, which is controlled by three semantic factors, namely, intrinsics, diversity, and randomness. A causal adversarial autoencoder (CAE) for SAR-ATR is then proposed to embody this disentangled representation, which incorporates a number of novel built-in network features. A physically reasonable cyclic high-frequency information-based embedding (CHE) method is proposed for azimuth encoding, which ensures the uniformity, continuity, periodicity, and distinctiveness of angle. A symmetrically conditional encoding (SCE) module is established to constrain the semantic consistency of low-dimensional features. Besides, a hybrid loss function is designed, which is composed of latent adversarial loss, reconstruction loss, and task-oriented losses. Both representation and generalization abilities are thoroughly evaluated through qualitative visualization and quantitative comparison experiments on the moving and stationary target acquisition and recognition (MSTAR) and FUSAR-Ship datasets. Experimental results demonstrate superior representation ability for the disentangled properties via angle interpolation and target transformation of SAR images. Using only 12 samples per class, the proposed CAE can achieve an accuracy of 93.1% for the ten-target SAR-ATR classification task.**

***Index Terms*— Causal adversarial autoencoder (CAE), disentangled representation, low-dimensional manifold, synthetic aperture radar (SAR), target recognition.**

I. Introduction

SYNTHETIC aperture radar (SAR) data are evolving toward higher resolution (HR), multidimension (MD), and multimodality (MM) directions, which accelerate the wide applications of intelligent algorithms in SAR-automatic target recognition (SAR-ATR) field [1]. Deep learning (DL) has revolutionized the field of computer vision (CV), and it has been demonstrated that DL-based SAR-ATR methods significantly outperform conventional methods in standard benchmark SAR-ATR datasets [2]. DL-based methods for SAR-ATR basically use three strategies: network structure modification [3], learning algorithm improvement [4], and image information enhancement [5]. The structure-modification-based methods improve the feature extraction ability by designing task-driven networks, e.g., combining different network layers and introducing various attention mechanisms. The second strategy is to improve the learning algorithm, which often takes into account the distribution discrepancies between optical and SAR sensors caused by different imaging mechanisms. The third strategy is to enhance image feature, i.e., scattering-information-based augmentation methods which mainly focus on the modeling of unique electromagnetic (EM) scattering characteristics of targets so as to optimize the decision boundary of the network. Note that most of these DL-based methods have achieved state-of-the-art performances in SAR-ATR.

However, different from optical images, SAR images reflect the EM characteristics of target with inherent speckle noise and unique geometric distortions, which are difficult to interpret intuitively [2]. Moreover, SAR imaging results are extremely sensitive to observation parameters which makes the problem even worse with very sparse training sample. The two major challenges now facing DL-based SAR-ATR are as follows.

1) *Lack of Interpretability:* The internal decision logic of DL is still in a "Black-Box" state. Besides, these data-driven models trained in CV may not be directly transferable to SAR images, as they do not fully consider the physical, statistical, and semantic characteristics that are unique to SAR images. Thus, the lack of interpretability makes it vulnerable and unreliable in decision-making, seriously restricting its practical application of SAR-ATR.
2) *Weak Generalization Ability:* Due to the acquisition cost of raw data, labor-intensive procedure, and limitation of sensor parameters, the number of target SAR samples is usually orders of magnitude fewer than the sample size of nature images [6], [7]. Furthermore, SAR images are susceptible to distortions related to target configurations and observation parameters due to the scattering scintillation effect [8]. Therefore,

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generalization of DL-based method with scarce training samples remains a challenging problem.

Recently, there has been an increasing interest in few-shot learning (FSL), which aims to improve generalization ability to new tasks with limited experience [9]. Typical FSL methods in CV field are also widely applied in SAR-ATR, which can be mainly categorized into three perspectives: data-oriented, model-oriented, and algorithm-oriented methods.

1) The data-oriented methods use prior information to augment training data. Relevant work for SAR data augmentation includes handcrafted rules, electromagnetic simulation, and deep generation. Handcrafted rule-based methods are implemented by image preprocessing transformations, e.g., translating, flipping, rotating, filtering, noising, and quantifying the training samples [10]. Another unique strategy in SAR, electromagnetic simulation based on computer-aided drawing (CAD) models, also plays an important role [11], [12]. An EM simulation-aided zero-shot learning method is proposed for SAR-ATR using EM simulated samples as training samples [13]. An alternative way to generate synthetic SAR images is to use deep generation models to generate samples directly [14]. Guo et al. attempt generating SAR samples using generative adversarial nets (GANs) [15]. The Wasserstein GAN is also adopted to generate new samples, which obtain a 79.59% recognition rate for ten-target recognition with 360 training samples [16]. Especially, the deep generative model was used to construct a continuous SAR target feature space for zero-shot learning [17].
2) Model-oriented methods use prior knowledge to constrain the hypothesis space, i.e., the neural network model, thereby reducing model overfitting risk. Transfer-learning-based methods extract general features across different domains, while metric-based methods conduct learning in a similarity-based contrast manner [18], [19]. Wang et al. [20] combined the transfer learning with the single-shot multibox detector to realize target detection in SAR images.
3) In addition, algorithm-oriented methods design the search strategy for the parameter of optimal hypothesis by prior knowledge. Meta-learning-based methods accomplish learning by accumulating proper initialization information [21], [22]. Yang et al. [23] designed the hierarchical embedding network with center calibration (HENC) in a meta-learning learning way to extract multiscale features for few-shot fine-grained SAR target classification. In particular, recent developments in FSL have also highlighted the need for interpretable models [24]. The transferability of deep features between natural and SAR images was investigated [25]. Song et al. proposed an autoencoder model to learn SAR image representation which incorporates physics-aware mechanism such as rotated cropping to address the representation of target orientation [26]. The existing methods mainly concentrate on the design of networks, generation strategies, and learning algorithms but fail to take the prior experience of targets into account. In addition, the models are not designed in an interpretable manner from the beginning. The combination of causality and DL has attracted increasing interest of CV community and has made impressive progress in many visual tasks, such as image synthesis and zero-shot recognition [27], [28]. CausalGAN [29] used adversarial training to learn causal implicit generation models and generated new images conditioned on different label combinations. CauaslVAE [30] learned disentangled representations of causal concepts and automatically discovers causal relationships between them with weakly supervised signals.

Inspired by the success of causal DL, a novel causal adversarial autoencoder (CAE) for SAR-ATR is proposed for SAR target representation with limited samples. Specifically, we first investigate a causal model of SAR image generation process and then propose a new representation of SAR manifold where the semantic factors are disentangled into three factors, namely, intrinsic, diversity, and randomness. Following the causal relationships of the disentangled factors, the CAE is constructed for SAR representation learning. The CAE uses an adversarial autoencoder structure where randomly sampled state is fed into a causal decoder together with the disentangled factors. The decoder learns to synthesize SAR images which are compared against real SAR images via the encoder followed by an adversarially trained discriminator. Physically reasonable encoding schemes are devised for each disentangled factors, respectively. For example, cyclic high-frequency information-based embedding (CHE) method is proposed for azimuth angle encoding. Moreover, hybrid loss function is used to train the CAE. Qualitative visualizations and quantitative comparisons are conducted on the moving and stationary target acquisition and recognition (MSTAR) dataset to demonstrate the effectiveness and superiority of the proposed model. Especially, the disentangled intrinsic properties are verified by angle-interpolation and target-transformation effects. The efficacy of diversity and randomness are evaluated through ablation experiments.

The remainder of this article is organized as follows: Section II describes the theoretic framework and algorithm implementation of the proposed CAE. The experimental results and performance evaluations are presented in Section III. Section IV summarizes this article with discussions.

II. Causal Graph Model for SAR Image Representation

A. Manifold Distribution Hypothesis of SAR Data

Neural data distribution hypotheses boil down to: the law of manifold distribution—high-dimensional natural data of the same class are concentrated near a low-dimensional manifold; and the law of clustering distribution—different subclasses correspond to different probability distributions on the manifold, where the distances between these probability distributions are large enough for classification [31], [32]. Deep neural networks (DNN) discover and learn the low-dimensional manifold through multiple nonlinear mappings. The same type of target will be formed into multiple

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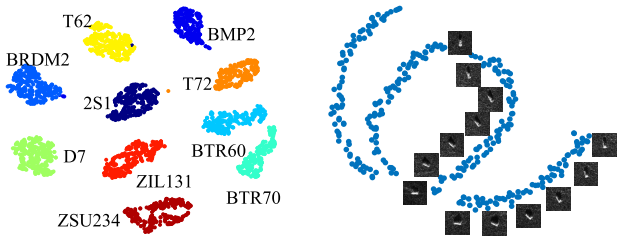


Fig. 1. Manifold distributions of the MSTAR dataset (15° depression angle). (a) Embedded results of ten categories of target. (b) Embedded results of an identified category of target.

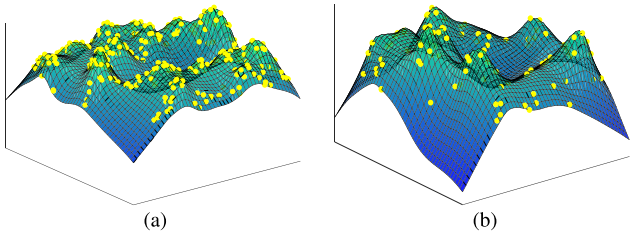


Fig. 2. Effects of few samples on data manifold distribution. Three-dimensional visualization of data manifold obtained with (a) large-scale samples and (b) few samples.

SAR images with different wavelengths, depression angles, azimuth angles of target, and backgrounds. With the change in observation conditions, continuously changing images satisfy certain constraints on the geometric distribution. Therefore, SAR images of a specific type can be regarded as a low-dimensional manifold embedded in high-dimensional space. According to the appearance frequency of images of different subclasses, it can be assumed that there is a certain probability distribution on the manifold.

Samples of the MSTAR dataset are adopted as examples to illustrate the rationality of the hypothesis [33], whose manifold is shown in Fig. 1. The dimension of image is 128×128. Therefore, an image can be seen as a point in $\mathbb{R}^{16384}$ space. A gold-standard network is trained with large-scale samples in advance, which is considered to contain sufficient prior information. Then, it is used to extract features of samples. The t-distributed stochastic neighbor embedding (t-SNE) is used to map the extracted features onto a planar region [34], where the data are s considered as concentrated around a low-dimensional manifold with each image corresponding to a point. Fig. 1(a) shows the low-dimensional manifold of ten targets, where images of the same category are mapped to the same cluster with a specific color. Fig. 1(b) visualizes the low-dimensional manifold of an identified category, demonstrating that the sample distribution varies regularly with the azimuth angle of target. Besides, samples with adjacent angles locate close to each other on the low-dimensional manifold.

B. FSL Problem Setup Under the Manifold Framework

FSL is a branch of machine learning problems defined by experience E, task T, and evaluation criterion P, where E contains only a little supervised information for T [9]. An error exists between the prediction and ground truth, which can be decomposed into an approximation error $\varepsilon_{\text{app}}(\mathcal{H})$ and an estimation error $\varepsilon_{\text{est}}(\mathcal{H})$, as shown in the following equations:

$$\mathbb{E}[\mathrm{R}(h_I) - \mathrm{R}(\hat{h})] = \mathbb{E}[\mathrm{R}(h^*) - \mathrm{R}(\hat{h})] + \mathbb{E}[\mathrm{R}(h_I) - \mathrm{R}(h^*)] \tag{1}$$

$$\varepsilon_{\text{app}}(\mathcal{H}) = \mathbb{E}[\mathrm{R}(h^*) - \mathrm{R}(\widehat{h})] \tag{2}$$

$$\varepsilon_{\text{est}}(\mathcal{H}, I) = \mathbb{E}[\mathrm{R}(h_I) - \mathrm{R}(h^*)] \tag{3}$$

where h^*, $\widehat{h}$, and h_I represent the ideal hypothesis, the optimal hypothesis in the existing hypothesis space $\mathcal{H}$, and the hypothesis obtained by the model, respectively. R is the expected error, and I is the number of training samples. I is small in FSL, which makes the assumption h_I that achieves the minimum empirical risk $\mathrm{R}(h_I)$ may be an assumption that deviates from h^*. As given by (1), the total error is determined by the size of hypothesis space $\mathcal{H}$ and the number of training samples I. Besides, Bayesian decision theory proves that no matter what kind of estimation $f(x)$ is used to approximate $\hat{f}$, the following equation is satisfied [35]:

$$\sup_{f \in \mathbb{F}} \mathbb{E}(\hat{f} - f(x))^2 \geq \text{const} \cdot N^{-2/(2+D)} \tag{4}$$

where N and D represent the number of training samples and the data dimension, respectively. $\mathbb{F}$ represents the set of all Lipschitz functions on $[0, 1]^D$. The sampling capacity required for modeling is closely related to the data dimension within a certain error limit. It also has an exponentially increasing relationship, which is named "The curse of dimensionality."

To avoid "The curse of dimensionality," it is considered to represent the data in a lower dimension, with maintaining effective information as much as possible. Therefore, the FSL problem is investigated under the manifold distribution hypothesis discussed in Section II-A. Visualizations of the low-dimensional manifolds obtained under large-scale samples and few samples are shown in Fig. 2, where the yellow point and the color of surface represent the sample and sampling density, respectively. The brighter the color, the denser the sampling probability. The denser the sampling density of data, the more explicit the obtained manifold. Besides, the tangent directions reveal possible distributions of data, while the normal directions represent the diversity. However, when only a few training samples are available, the data distribution is sparser with divergent structure.

The effect of the azimuth-sparse training samples on the low-dimensional manifold distribution of the data can be significant and could potentially limit the ability of a model to generalize to unseen azimuth angles, which can be summarized into three aspects.

1) *Limited Coverage:* If the training data only cover a limited range of azimuth angles, the learned low-dimensional manifold might also only cover a corresponding limited region of the manifold space. This could mean that the model might not perform well when asked to generalize to unseen angles.
2) *Biased Representation:* The learned low-dimensional manifold might be biased toward the angles present in the training data. This could distort the manifold structure and lead to poor performance when the model is used on data from other angles.
3) *Incompleteness:* The manifold might not accurately represent the full complexity and variability of the data if it

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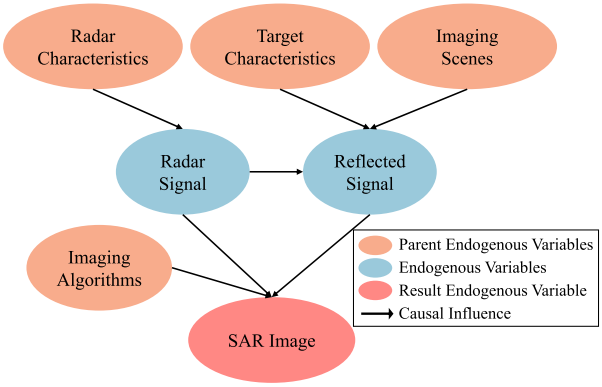


Fig. 3. Causal graph for SAR image generation process.

is learned from azimuth-sparse training samples. Certain features or variations that only appear at certain angles might not be captured at all.

Therefore, prior probabilities available before training are used to constrain the optimization procedure. Prior probability distributions are injected with training data to reduce the sample complexity required by the model, thereby reducing the uncertainty of empirical risk minimization under FSL condition.

C. Causal Model for Disentangled Representation of SAR Image

The effectiveness of neural networks lies in the ability of disentangling and flattening data manifolds [36]. A set of high-dimensional SAR images can be regarded as a low-dimensional manifold, which is controlled by fewer factors. From a causally inspired perspective, we aim to model the causal relationship between these disentangled factors and learn the causal disentangled representation of SAR images, which can lead to better generalization from small amounts of data. Besides, through the causal disentangled representation, prior distributions of influencing factors, including the category and azimuth, can be obtained and modeled in advance, thereby constraining the sampling and optimization of the deep model.

The causal graph of the imaging process of SAR can be modeled as a complex network of variables $V = \{X_1, X_2, \ldots, X_n\}$ and functions $F = \{f_1, f_2, \ldots, f_n\}$, which implicit different exogenous variables and imaging mechanism of SAR images [37]. X causes Y which can be formulated as $Y = f(X, E)$ or $X \to Y$. This means that the relationship f can map X to Y with several variables E independent of X. E is also named exogenous variables, which are determined outside the system [29], [30]. As shown in Fig. 3, a basic causal graph for the image generation process of SAR is established, which includes the following factors and relationships. The properties of the radar, such as frequency, power, and depression angle, as well as the characteristics of the target, including its category and azimuth angle, and the conditions of the environment and the imaging algorithms used, are some of the key external factors that can influence the quality of SAR images. The radar signal is a confounder, because it affects both the reflected signal and SAR image. However, when constructing the proposed CAE model to implement the disentangled representation, the causal relationship between the radar signal, reflected signal, and SAR image is implicitly learned and represented through the neural network. These variables and relationships can be further expanded and refined to represent a more complete causal graph of the actual SAR process.

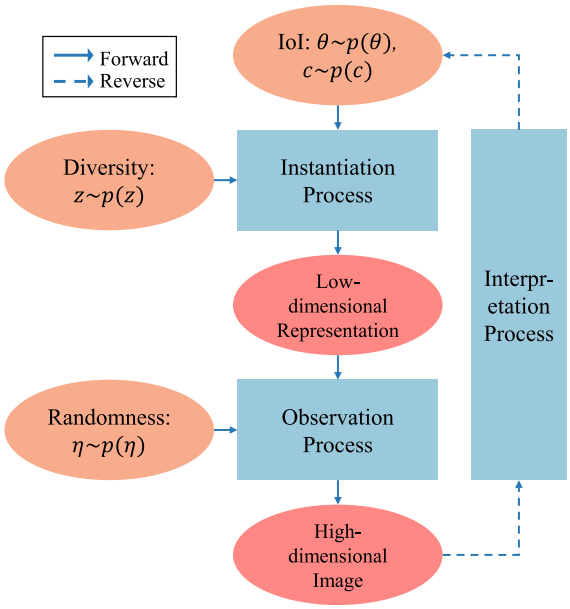


Fig. 4. Causal modeling for disentangled representation of SAR images. Orange ellipse denotes exogenous variable that follows prior distributions, red ellipse denotes endogenous variable, and blue rectangle denotes the mapping function.

Inspired by the proposed causal graph as shown in Fig. 3, an instantiated diagram that includes the forward imaging process and the reverse interpretation process of SAR images is further proposed. To simplify the modeling of neural network, a disentangled representation for SAR images is constructed based on the proposed causal model of SAR images. Disentangled features are assumed to locate in a flatter latent space, which produces smoother representational transitions than a highly curved space. As shown in Fig. 4, it mainly consists of three semantic factors: intrinsic, diversity, and randomness, which summarize the endogenous factors in Fig. 3. Specifically, the intrinsic information of interest (IoI) mainly concentrated on the semantic characteristics, including azimuth angle and category of target, which is considered to be of interest to ATR. In addition to IoI, the noninterest information, including diversity and randomness, is also indispensable for representing SAR images. Due to its unique imaging mechanism, SAR images are extremely sensitive to observation parameters. On one hand, there always exists inherent speckle noise and scintillation effect caused by coherent processing and movement of stationary phase points, which is specified as the randomness of SAR images. On the other hand, affected by target deformation configurations and observation scenes, imaging results of the same target are also with slight fluctuations, which is referred to as diversity.

Instantiation and observation processes represent different causal mechanisms, such as how randomness, diversity, and low-dimensional representations jointly determine the generated SAR images. The interpretation process represents the abduction process (inferring the cause from the effect),

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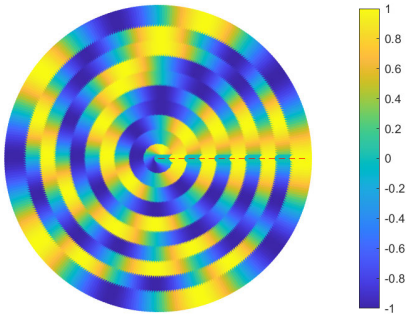


Fig. 5. Visualization of CHE (the red dashed line indicates encoding result of 0/360°).

i.e., given an SAR image, inferring its category and azimuth. Different from general disentangled probabilistic graphical models, causal models learn hierarchical causal relationships between variables, from the diversity to low-dimensional representation, and then to images. In addition, it can answer "what if" questions, i.e., manipulate variables and predict the outcome of interventions [30]. For example, the proposed CAE can be implemented to the effects of angle interpolation, target transformation, and the noninterest information representation on the generated images in a causal manner.

One-hot encoding is used to map the category to discrete, mutually exclusive, and unordered features. Different from category, angle is continuous and periodical. Besides, images of adjacent angles tend to be with high correlations. However, due to the translation-invariant characteristics of convolution layers, DNN is capable of representing translation, but not rotation. Thus, inspired by the position encoding in transformer [38], a CHE method is proposed for angle information encoding. The strategy of CHE is shown in the following equation:

$$r(\theta) = [\sin(\theta), \cos(\theta), \sin(2\theta), \cos(2\theta), \ldots, \sin(L\theta), \cos(L\theta)] \tag{5}$$

where $\theta \in [0, 2\pi]$. θ and $r(\theta)$ represent the original and encoded angles, respectively. L is set as 5 to compromise between feature dimension and effectiveness. Through combined trigonometric functions, the proposed CHE achieves dimension expansion of angle features. The value of encoded angle distributes uniformly in $[-1, 1]$, which is beneficial to the optimization of DNN. CHE not only ensures the uniformity, continuity, uniqueness, and periodicity of angle but also improves the distinctiveness of adjacent angles. Visualization of the CHE is given in Fig. 5, where the radial color coding represents the encoding vector.

In addition, multidimensional Gaussian distributions are introduced to represent the randomness and diversity of SAR images. Since the disentangled features are composed of known patterns and explicit distributions, they can be used as supervised information to facilitate complete pattern sampling and avoid pattern collapse effectively.

III. Causal Adversarial Autoencoder for SAR-ATR

A. Framework of the Proposed CAE

Guided by the disentangled representations of SAR images shown in Fig. 4, a causal adversarial autoencoder for SAR-ATR is proposed and the network structure is correspondingly designed, whose framework is presented in Fig. 6. It is established by separately modeling the intrinsic, diverse, and random features. The intrinsic properties correspond to the optimal boundaries of different subclasses, which are conducive to classification and prediction. The diverse and random terms correspond to the perturbation of images, which improves the generalization ability. Inspired by the adversarial latent autoencoder [39], optimization in low-dimensional space is adopted to constrain the size of hypothesis space, alleviating the curse of dimensionality under FSL conditions. Besides, prior distributions of SAR data are introduced to augment supervised information.

The CAE achieves manifold learning and probability distribution transformation through adversarial learning among four subnetworks, including mapper $\mathcal{M}$, generator $\mathcal{G}$, encoder $\mathcal{E}$, and discriminator $\mathcal{D}$. First, $\mathcal{M}$ transforms the prior probability distributions into the low-dimensional probability distributions of the latent space. Then, the decoding and encoding maps between the latent space representation and the image manifold are computed by $\mathcal{G}$ and $\mathcal{E}$, respectively. Finally, $\mathcal{D}$ computes the distance between the prior low-dimensional distribution and the generated low-dimensional distribution. Prior distributions, including the azimuth angle $\theta \sim p(\theta) = U(0, 2\pi)$, category $c \sim p(c) = \text{cat}(c)$, diversity $z \sim p(z) = \mathcal{N}(0, 1)$, and randomness $\eta \sim p(\eta) = \mathcal{N}(0, 1)$, are injected into different stages of CAE in a causally interpretable way, being consistent with Fig. 4. Specifically, z is first used to represent the diversity caused by the variations in target configurations and observation scenes, which are separate from intrinsic information. Then, θ and c are introduced into the predisentangled low-dimensional representation μ transformed from z. η is used in each generator block of $\mathcal{G}$ to simulate the perturbation generated during the SAR observation process.

B. Loss Functions and Implementations

CAE transforms the optimization from the high-dimensional image space to low-dimensional feature space through the adversarial learning between latent vectors. Consequently, it can be applied to multiple tasks, including image reconstruction, image generation, target classification, angle prediction, etc. The implementation process mainly consists of four parts. First, $\mathcal{M}$ is used to transform the prior Gaussian distribution $z \sim p(z) = \mathcal{N}(0, 1)$ into a preliminary low-dimensional distribution $q_{\mathcal{M}}(\mu)$

$$q_{\mathcal{M}}(\mu) = \int_z q(\mu|z)p(z)dz. \tag{6}$$

Multiple influence factors are entangled in the initial low-dimensional distributions obtained from data. Although some of them can be modeled by semantics attributes, there still exists noninterest information that relies on neural networks for representation. Moreover, no evidence yet demonstrates that it satisfies common probability distributions, such as Gaussian and uniform distributions. Therefore, $\mathcal{M}$ is applied first to reduce the difficulty in the transformation from prior to a more proper representation probability distribution.

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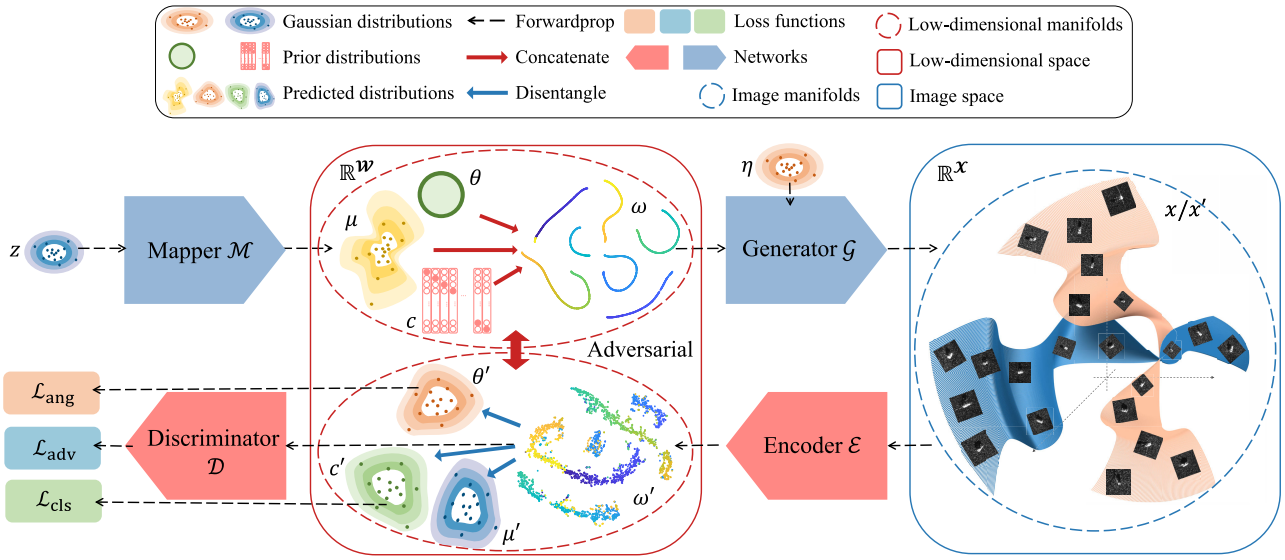


Fig. 6. Architecture of CAE, consisting of four subnetworks. Specifically, mapper $\mathcal{M}$ and generator $\mathcal{G}$ represent the instantiation process and observation process, respectively. The encoder $\mathcal{E}$ and discriminator $\mathcal{D}$ represent the reverse interpretation process. Besides, z denotes the diversity, η denotes the randomness, angle θ and category c jointly represent the IoI, and ω denotes the low-dimensional representation.

Then, θ, c, and μ are concatenated as a low-dimensional representation vector ω

$$q(\omega) = \int_u \int_c \int_\theta q(\omega|\mu, \theta, c) q_M(\mu) p(c) p(\theta) d\theta dc d\mu. \quad (7)$$

Consequently, the generator $\mathcal{G}$ performs conditional sample generation and reconstruction based on the low-dimensional representation vector ω

$$q_{\mathcal{G}}(x') = \int_\omega \int_\eta q(x'|\omega, \eta) q(\omega) p(\eta) d\eta d\omega. \quad (8)$$

Furthermore, the encoder $\mathcal{E}$ encodes the real image x or fake image x' into a low-dimensional representation ω', remapping manifolds from the image space $\mathbb{R}^{\mathcal{X}}$ to the latent space $\mathbb{R}^{\mathcal{W}}$. For real training images, the posterior distribution of ω' is

$$q_{\mathcal{E}}^{\text{real}}(\omega') = \int_x q(\omega'|x) p_{\text{data}}(x) dx \quad (9)$$

where $x \sim p_{\text{data}}$ is the Dirac distribution of real samples. On the contrary, the posterior distribution of ω' for fake images satisfies

$$q_{\mathcal{E}}^{\text{fake}}(\omega') = \int_x q(\omega'|x) q_{\mathcal{G}}(x) dx. \quad (10)$$

The prior low-dimensional vector ω has semantic consistency with the encoded low-dimensional vector ω' in the corresponding positions. That is, predicted labels and angles can be directly obtained through the low-dimensional representation vector ω'

$$q_{\mathcal{E}}(\theta') = \int_{\omega'} q(\theta'|\omega') q_{\mathcal{E}}(\omega') d\omega' \quad (11)$$

$$q_{\mathcal{E}}(c') = \int_{\omega'} q(c'|\omega') q_{\mathcal{E}}(\omega') d\omega'. \quad (12)$$

Finally, $\mathcal{D}$ is designed to discriminate the reliability and conditional consistency of the low-dimensional representation vectors. Ensure the low-dimensional distribution encoded by $\mathcal{E}$ be consistent with the prior low-dimensional distribution

$$q_{\mathcal{E}}(\omega') = q(\omega). \quad (13)$$

It essentially constrains the image distributions to be the same

$$q_{\mathcal{G}}(x') = p_{\text{data}}(x). \quad (14)$$

For this purpose, adversarial learning loss $\mathcal{L}_{\text{adv}}$ and reconstruction loss $\mathcal{L}_{\text{rec}}$ are considered

$$\begin{aligned}\mathcal{L}_{\text{adv}} &= \min_{\mathcal{M},\mathcal{G}} \max_{\mathcal{E},\mathcal{D}} (f(\mathcal{D} \circ \mathcal{E}) + f(-\mathcal{D} \circ \mathcal{E} \circ \mathcal{G} \circ \mathcal{M})) \\ &= \min_{\mathcal{M},\mathcal{G}} \max_{\mathcal{E},\mathcal{D}} \big(\mathbb{E}_{x \sim p_{\text{data}}(x)}[f(\mathcal{D} \circ \mathcal{E}(x))] \\ &\quad + \mathbb{E}_{z \sim p(z), \theta \sim p(\theta), c \sim p(c)}[f(-\mathcal{D} \circ \mathcal{E} \circ (\mathcal{G} \circ (\mathcal{M}(z), \theta, c))]\big). \end{aligned} \quad (15)$$

$$\begin{aligned}\mathcal{L}_{\text{rec}} &= \min_{\mathcal{E},\mathcal{G},\mathcal{M}} \Delta(\mathcal{M} \| \mathcal{E} \circ \mathcal{G} \circ \mathcal{M}) \\ &= \min_{\mathcal{E},\mathcal{G},\mathcal{M}} \mathbb{E}_{z \sim p(z), \theta \sim p(\theta), c \sim p(c)} \Delta((\mathcal{M}(z), \theta, c) \| \\ &\quad (\mathcal{E} \circ \mathcal{G} \circ (\mathcal{M}(z), \theta, c))) \end{aligned} \quad (16)$$

where $f(t) = t$. Besides, $\Delta(p\|q)$ represents the measure of discrepancy between p and q, which is calculated by the L2-norm in experiments. Therefore, the key idea of CAE is fundamentally different. It does not require the low-dimensional encoding distribution to approximate any known distributions but optimizes it through a self-constrained style. In addition, the task-oriented losses are further optimized in a synergistic learning style, including the classification and prediction of angle

$$\begin{aligned}\mathcal{L}_{\text{angd}} = \min_{\mathcal{E}} \Big(&\mathbb{E}_{x \sim p_{\text{data}}(x), \theta_{\text{real}} \sim p_\theta^{\text{data}}(\theta)} \left[f_{\text{ang}}(\mathcal{E}(x)|\theta_{\text{real}}) \right] \\ &+ \mathbb{E}_{\theta \sim p(\theta)} \left[f_{\text{ang}}(\mathcal{E} \circ \mathcal{G}(\theta)|\theta) \right] \\ &- \mathbb{E}_{x \sim p_{\text{data}}(x), \theta_{\text{real}} \sim p_\theta^{\text{data}}(\theta), \theta \sim p(\theta)} \left[f_{\text{ang}}(\mathcal{E}(x)|\theta, \theta \neq \theta_{\text{real}}) \right] \\ &- \mathbb{E}_{\theta, \theta_{\text{fake}} \sim p(\theta)} \left[f_{\text{ang}}(\mathcal{E} \circ \mathcal{G}(\theta_{\text{fake}})|\theta, \theta \neq \theta_{\text{fake}}) \right] \Big) \end{aligned} \quad (17)$$

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$$\mathcal{L}_{\text{angg}} = \min_{\mathcal{G}} \mathbb{E}_{\theta\sim p(\theta)}\left[f_{\text{ang}}(\mathcal{E}\circ\mathcal{G}(\theta)|\theta)\right] \tag{18}$$

$$\begin{aligned}\mathcal{L}_{\text{clsd}} = \min_{\mathcal{E}}\big(&\mathbb{E}_{x\sim p_{\text{data}}(x),c_{\text{real}}\sim p_c^{\text{data}}(c)}\left[f_{\text{cls}}(\mathcal{E}(x)|c_{\text{real}})\right]\\ &+\mathbb{E}_{c\sim p(c)}\left[f_{\text{cls}}(\mathcal{E}\circ\mathcal{G}(c)|c)\right]\\ &-\mathbb{E}_{x\sim p_{\text{data}}(x),c_{\text{real}}\sim p_c^{\text{data}}(c),c\sim p(c)}\left[f_{\text{cls}}(\mathcal{E}(x)|c\neq c_{\text{real}})\right]\\ &-\mathbb{E}_{c,c_{\text{fake}}\sim p(c)}\left[f_{\text{cls}}(\mathcal{E}\circ\mathcal{G}(c_{\text{fake}})|c,c\neq c_{\text{fake}})\right]\big)\end{aligned} \tag{19}$$

$$\mathcal{L}_{\text{clsg}} = \min_{\mathcal{G}} \mathbb{E}_{c\sim p(c)}\left[f_{\text{cls}}(\mathcal{E}\circ\mathcal{G}(c)|c)\right]. \tag{20}$$

The CAE accomplishes classification and prediction through setting task-oriented adversarial losses without introducing extra networks. f_{ang} and f_{cls} are the L2-norm and cross-entropy losses, respectively. θ_{real} and c_{real} represent the corresponding angle and category, respectively, while θ_{fake} and c_{fake} are the corresponding ground truth for fake images. CAE is trained alternately by adversarial low-dimensional representations in the latent space, and detailed implementation is given in Algorithm 1.

Algorithm 1 CAE Algorithm

Input: Training samples $D_{\text{train}} = \{x_i, \theta^i_{\text{real}}, c^i_{\text{real}}\}_{i=1}^n$, learning rates $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ and batch size n_{bs}

Initialize model parameters $\Theta_{\mathcal{M}}$, $\Theta_{\mathcal{G}}$, $\Theta_{\mathcal{E}}$, and $\Theta_{\mathcal{D}}$
Randomly sample batches: $\{x_i, \theta^i_{\text{real}}, c^i_{\text{real}}\}_{i=1}^{n_{\text{bs}}} \sim p_{\text{data}}(x)$, $\{z_i\}_{i=1}^{n_{\text{bs}}} \sim p(z)$, $\{\theta^i_{\text{fake}}\}_{i=1}^{n_{\text{bs}}} \sim p(\theta)$, $\{c^i_{\text{fake}}\}_{i=1}^{n_{\text{bs}}} \sim p(c)$
while *not converged* **do**
  for *each batch of size* n_{bs} **do**
    Calculate $\mathcal{L}_{\text{adv}}$ by (15)
    Let $\Theta_{\text{advd}} = \{\Theta_{\mathcal{E}}, \Theta_{\mathcal{D}}\}$ and Update $\Theta_{\text{advd}} \leftarrow \text{Adam}\left(\nabla_{\Theta_{\text{advd}}}\mathcal{L}_{\text{adv}}(\Theta_{\text{advd}}), \Theta_{\text{advd}}, \alpha_1\right)$
    Let $\Theta_{\text{advg}} = \{\Theta_{\mathcal{M}}, \Theta_{\mathcal{G}}\}$ and Update $\Theta_{\text{advg}} \leftarrow \text{Adam}\left(\nabla_{\Theta_{\text{advg}}}\mathcal{L}_{\text{adv}}\left(\Theta_{\text{advg}}\right), \Theta_{\text{advg}}, \alpha_1\right)$
  end
  for *each batch of size* n_{bs} **do**
    Calculate $\mathcal{L}_{\text{rec}}$ by (16)
    Let $\Theta_{\text{rec}} = \{\Theta_{\mathcal{E}}, \Theta_{\mathcal{G}}, \Theta_{\mathcal{M}}\}$ and Update $\Theta_{\text{rec}} \leftarrow \text{Adam}\left(\nabla_{\Theta_{\text{rec}}}\mathcal{L}_{\text{rec}}(\Theta_{\text{rec}}), \Theta_{\text{rec}}, \alpha_2\right)$
  end
  for *each batch of size* n_{bs} **do**
    Calculate $\mathcal{L}_{\text{angd}}$ by (17)
    Update $\Theta_{\mathcal{E}} \leftarrow \text{Adam}\left(\nabla_{\Theta_{\mathcal{E}}}\mathcal{L}_{\text{angd}}(\Theta_{\mathcal{E}}), \Theta_{\mathcal{E}}, \alpha_3\right)$
    Calculate $\mathcal{L}_{\text{angg}}$ by (18)
    Update $\Theta_{\mathcal{G}} \leftarrow \text{Adam}\left(\nabla_{\Theta_{\mathcal{G}}}\mathcal{L}_{\text{angg}}(\Theta_{\mathcal{G}}), \Theta_{\mathcal{G}}, \alpha_3\right)$
  end
  for *each batch of size* n_{bs} **do**
    Calculate $\mathcal{L}_{\text{clsd}}$ by (19)
    Update $\Theta_{\mathcal{E}} \leftarrow \text{Adam}\left(\nabla_{\Theta_{\mathcal{E}}}\mathcal{L}_{\text{clsd}}(\Theta_{\mathcal{E}}), \Theta_{\mathcal{E}}, \alpha_4\right)$
    Calculate $\mathcal{L}_{\text{clsg}}$ by (20)
    Update $\Theta_{\mathcal{G}} \leftarrow \text{Adam}\left(\nabla_{\Theta_{\mathcal{G}}}\mathcal{L}_{\text{clsg}}(\Theta_{\mathcal{G}}), \Theta_{\mathcal{G}}, \alpha_4\right)$
  end
end
return $con(r_i)$;

C. SCE Module

Remarkably, a symmetrically conditional encoding (SCE) module is constructed to achieve the semantical consistence of low-dimensional representations. As shown in Fig. 7, the SCE module is composed of $\mathcal{E}$ and $\mathcal{G}$. Based on the specified low-dimensional vectors output by $\mathcal{M}$, the generator

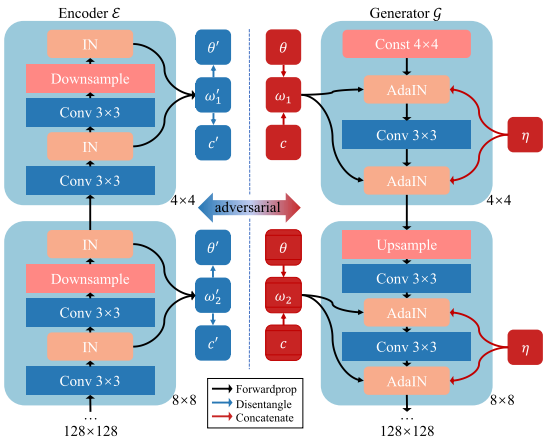


Fig. 7. Structure of the SCE module.

TABLE I
INFORMATION OF MSTAR EXPERIMENTAL DATASETS (17°)

Azimuth Intervals [°]	Training Samples	Test Samples
5	692	2054
10	351	2395
15	230	2516
30	176	2570
24	141	2605
30	121	2625
35	97	2649
40	89	2657

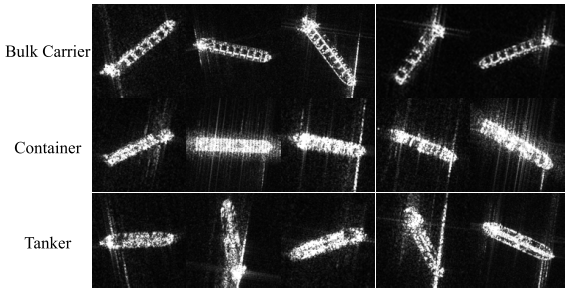


Fig. 8. Several training samples from the FUSAR-Ship dataset.

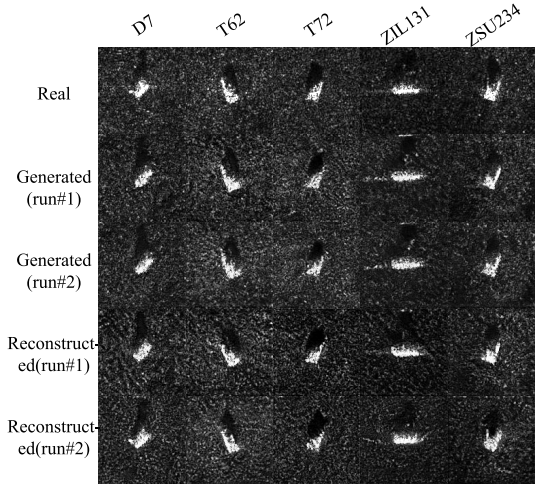


Fig. 9. Illustrations of the real, generated, and reconstructed samples of MSTAR.

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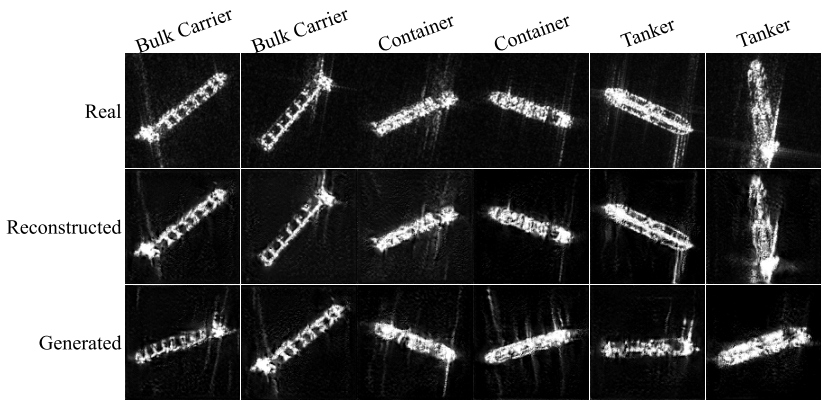


Fig. 10. Representation performance of the CAE on ships of Gaofen-3 satellite imagery.

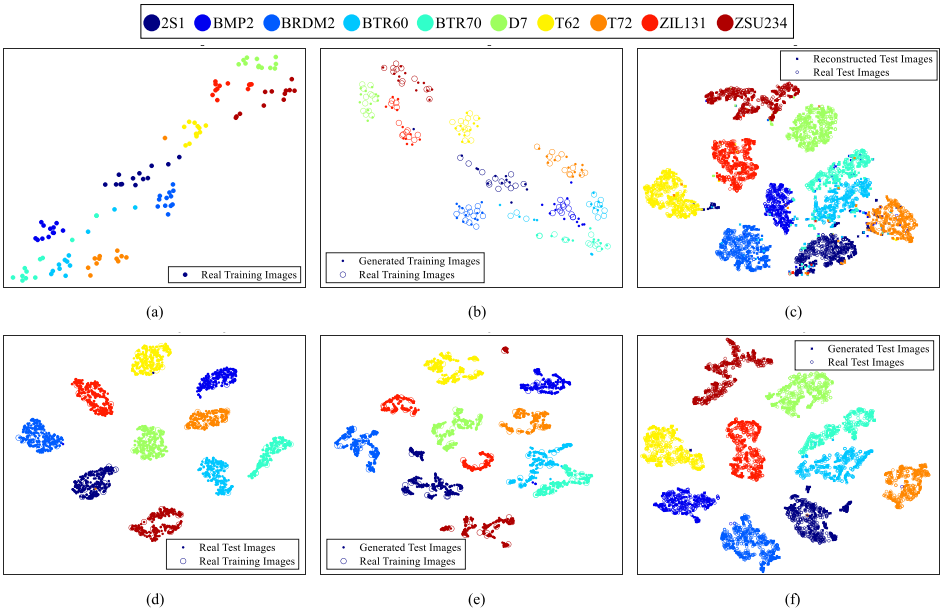


Fig. 11. Visualizations of fake and real samples using t-SNE. (a) Real training images. (b) Generated training images vs. real training images. (c) Reconstructed test images vs. real test images. (d) Real test images vs. real training images. (e) Generated test images vs. real training images. (f) Generated test images vs. real test images.

progressively generates 128×128 images from 4×4 constant values. The AdaIN and IN are used to extract styles which are conducive to representation [39], [40]. Because of the symmetrical structures of $\mathcal{E}$ and $\mathcal{G}$, the predicted angle and category of the target can be disentangled directly from fixed positions of vectors encoded by $\mathcal{E}$.

IV. Experiments and Evaluations

A. Dataset Description and Experimental Implementation

1) Experimental Datasets: Experiments are conducted on the MSTAR [33] and FUAR-Ship datasets [41]. MSTAR contains ten types of vehicles at various viewing angles with a resolution of 0.3×0.3 m, and the chip size is 128×128 pixel. The azimuth angle is 0°–360°, with an interval of about 1°–2°. The depression angles are 17° and 15°. To evaluate the effectiveness of CAE under FSL conditions, the training and test sets are randomly divided according to different azimuth angle intervals of training samples. The number of training samples of different categories under the same interval is evenly distributed. As the interval of training samples increased from 5° to 40°, the number of training samples decreased from 692 to 89. Specifically, when the angle interval is 30°, there are 121 training samples with 17° depression angle, and 2625 and 2426 test samples with 17° and 15° depression angles, respectively. The detailed number of training and test samples with 17° corresponding to different azimuth angle intervals is shown in Table I.

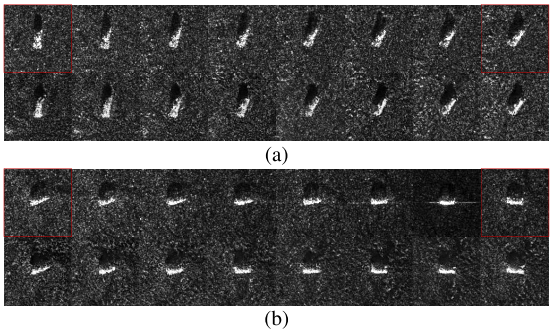


Fig. 12. Illustrations of azimuth angle interpolation. The top and bottom rows are real and generated samples, respectively. Training samples are highlighted with red boxes. (a) 2S1. (b) BRDM2.

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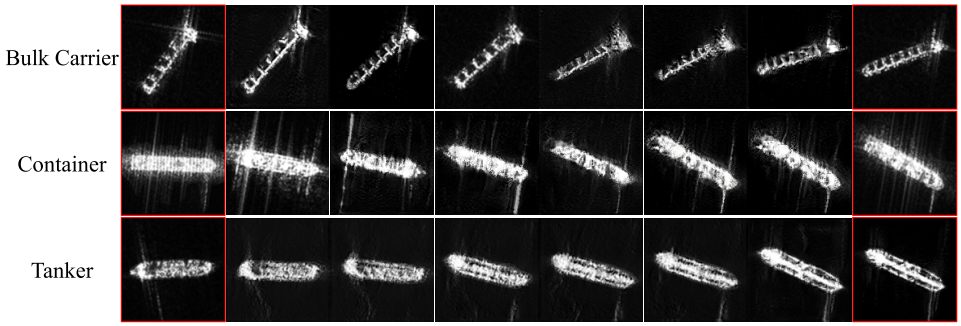


Fig. 13. Illustrations of azimuth angle interpolation of ship. Training samples are highlighted with red boxes.

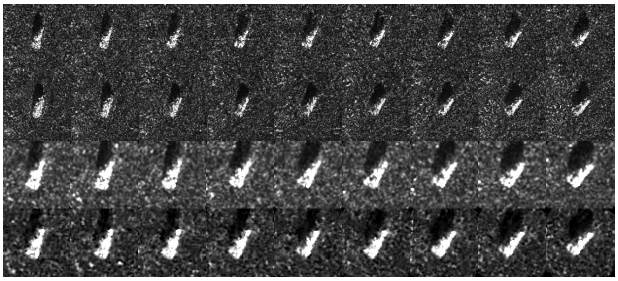


Fig. 14. Comparisons between the proposed CAE and AAE [26]. From top to bottom rows are real samples of CAE, generated samples of CAE, real samples of AAE, and generated samples of AAE, respectively.

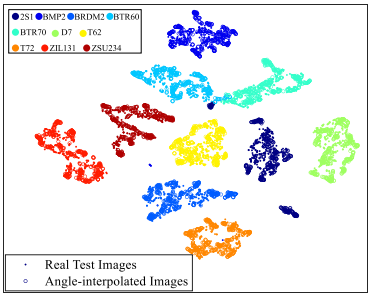


Fig. 15. Visualization of angle interpolation using t-SNE.

In addition, three typical ships of the FUSAR-Ship dataset in Gaofen-3 SAR images, including the bulk carrier, container, and tanker, are also selected as the experimental data to further evaluate the representation ability of CAE to other real-world targets. The imaging mode is sliding spotlight with a resolution of 1 m. A total of 20 slices from three categories are used as the training set, and several training samples are shown in Fig. 8. The azimuth distributions of ship training sample are sparser, and there also exists significant sidelobe effects. Affected by the complexity of ship structure and the diversity of background conditions, compared with the MSTAR dataset, the intraclass distance of the ship dataset is larger, and the distance between classes is smaller, which increases the difficulty in model representation.

2) Implementation Details: Experiments are implemented on the PyTorch framework, performed on NVIDIA GeForce RTX 2080 Ti. The dimension of low-dimensional representation is set as 64, where $[1, 10]$ and $[11, 10 + \dim(c)]$ dimensions are assigned angle and category attributes, respectively. The model is trained through six stages with different image resolutions, progressively growing from 4×4 to 128×128. Each stage is trained 100 epochs with $n_{\text{bs}} = [30, 30, 30, 30, 20, 15]$. The initial learning rates are set to $\alpha_1 = 1\text{e}-3, \alpha_2 = 1\text{e}-3, \alpha_3 = 9.5\text{e}-4$ and $\alpha_4 = 9.5\text{e}-4$. Four desired prior physical distributions are used as prior information, including the azimuth angle $\theta \sim p(\theta) = U(0, 2\pi)$, category $c \sim p(c) = \text{cat}(c)$, diversity $z \sim p(z) = \mathcal{N}(0, 1)$, and randomness $\eta \sim p(\eta) = \mathcal{N}(0, 1)$.

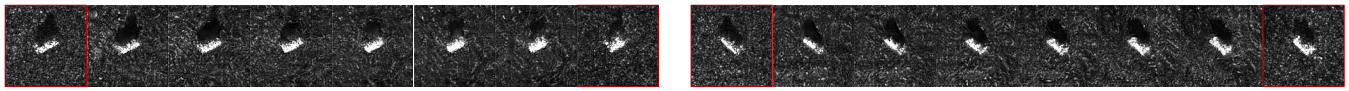
B. Evaluation of the Representation Performance

1) Generation and Reconstruction Ability: First, the CAE is trained with only 121 samples of ten targets to verify the representation performance of the proposed method. The trained model is then used to reconstruct and generate the training and test samples, respectively. Illustrations of real, generated, and reconstructed samples are presented in Fig. 9. Besides, two groups of the generated and reconstructed images are synthesized with the same intrinsic but different diversity and randomness attributes to show the representation of diverse samples. Targets are well-described in the generated image, while the background texture is varied. It is considered reasonable and effective because z and η are purposely injected to improve diversity and randomness.

Experiments are also conducted on the ship dataset to evaluate the representation performance. First, 20 samples are used to train the CAE, and the well-trained model is adopted to generate new samples. The experimental results of ship are shown in Fig. 10, where the top row is the training data, the middle row is the reconstructed training data, and the bottom row is the data generated with other angles. The experimental results indicate that the proposed model trained with few samples, despite slight blurring and distortion in sidelobe representation, has generally achieved realistic representation ability on both the reconstructing and generating images of other azimuth angles.

Furthermore, t-SNE is introduced to qualitatively analyze the distribution consistency between fake and real samples. The visualization results of the MSTAR dataset are shown in Fig. 11. As depicted in Fig. 11(a), when only training samples are available, the separability of the low-dimensional representations is limited. However, as the number of samples increases, the classification boundaries become more distinct. Fig. 11(b) verifies that the generated data distributions of different categories are aligned with real data

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(a) (b)

Fig. 16. Illustrations of target interpolation with increasing transformation rate [0, 1]. Real samples are highlighted with red boxes. (a) Transformation from 2S1 to D7 (azimuth angle = 60°). (b) Transformation from 2S1 to T62 (azimuth angle = 315°).

distributions, with slight deviations due to noninterest information. Fig. 11(e) demonstrates that the generated test data are primarily distributed in the area covered by the training samples, effectively achieving feature interpolation. The experimental results in Fig. 11(c) and (f) show that both the generated and reconstructed data are in close agreement with the real data distribution, with subcategories being mapped to the same regions. Moreover, the experimental results demonstrate that the generated data have a better distribution consistency and separability compared with the reconstructed data. This also validates the finding that more disentangled low-dimensional representations are beneficial for generation, as reported in previous studies [40].

2) Angle-Interpolation Ability: The disentangled performance of the angle attribute is further assessed through an angle-interpolation experiment. The experiment begins by encoding an initial sample of a fixed category using $\mathcal{E}$. Then, the azimuth attribute is gradually interpolated from the initial angle to the target angle, and $\mathcal{G}$ generates the corresponding samples based on the interpolated vectors. Fig. 12 presents two pairs of samples generated through angle interpolation with different categories of the MSTAR dataset, including azimuth ranges of [2°, 30°], and [67°, 96°].

Angle-interpolation experiments are also conducted on the ship dataset, whose results are presented in Fig. 13, with azimuth varying in [37°, 68°], [105°, 127°], and [262°, 293°]. Training samples are highlighted with red boxes. It is worth noting that since the FUSAR-Ship dataset does not contain samples of the same category with complete azimuth angles, there are no corresponding real samples to refer to for the samples generated by angle interpolation.

As evidenced by Figs. 12 and 13, the results of angle-interpolation experiment further highlight the disentangled performance of the angle attribute. The samples generated through the interpolation of the azimuth attribute show a satisfactory level of similarity to the target contour and consistency in angle with the real data. The model's overfitting to specific angles is effectively prevented by the proposed CAE, ensuring that the generated images rotate appropriately as the azimuth changes. Although generated images have achieved realistic generation performance, there still exist some subtle differences and blurriness compared with the real images. It is assumed primarily due to the injection of diversity and randomness representation terms and the inherent limitations of the generation model.

In addition, Fig. 14 provides a comparison between the proposed CAE and AAE [26]. The results demonstrate that CAE outperforms AAE in terms of the quality of angle interpolation, producing clearer target contours and more realistic scene textures without the need for contour maps as priors. These findings validate the effectiveness of the proposed CAE in learning the distribution of data based on a limited number of samples.

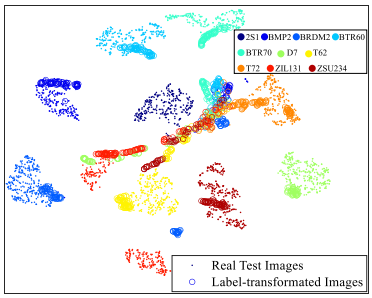


Fig. 17. Visualization of target-transformation effect using t-SNE.

Furthermore, t-SNE analysis is conducted to investigate the ability of the proposed CAE to maintain category distribution when generating images through angle interpolation. Fig. 15 provides a comparison between the angle interpolated data generated within [1°, 360°] and the test data. The generated data effectively populate the regions covered by the real data of the corresponding category. The experimental results demonstrate that the CAE can achieve azimuth generalization while preserving other intrinsic information, indicating its ability to capture the low-dimensional manifold of the data and perform effective angle interpolation.

3) Target-Transformation Ability: To further evaluate the disentangled performance of label attribute, the proposed CAE was used to conduct a target interpolation experiment. An image of the category 2S1 of a fixed angle was first encoded by $\mathcal{E}$ to obtain an initial representation. Then, the label attribute was linearly interpolated to obtain target representations, and $\mathcal{G}$ was used to generate images based on these representations. Two pairs of target-transformation result are presented in Fig. 16, where the leftmost, rightmost, and the middle parts represent the source, target, and images generated with a increasing transformation ratio $\gamma \in [0, 1]$, respectively. Taking Fig. 16(a) as an example, as γ increases, the target size in the generated image decreases, and the shadow changes accordingly, while the angle information remains 60°. Although these generated images correspond to objects that do not exist in the real world, the CAE is able to generate authentic images. These results demonstrate that the proposed CAE is capable of effectively disentangling the label attribute while preserving other intrinsic information.

In addition, visualizations of the target transformation using t-SNE are shown in Fig. 17, which presents a comparison between the distributions of test data and target-transformed data. The generated target-transformed images establish a connection between the source and target domains, with the distribution of the generated samples gradually spreading from the source area to the target area as γ increases. These transformations are believed to be continuous in the

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TABLE II
CLASSIFICATION RESULTS WITH ONLY 12 TRAINING SAMPLES PER CLASS

Algorithm	Test Accuracy (17°)	Test Accuracy (15°)	Time Complexity [sec/img]
AConvNet	68.08%	62.53%	**0.0057**
ShuffleNet	87.77%	85.41%	0.0136
EfficientNet	82.13%	81.78%	0.0184
SwinTransformer	85.49%	79.27%	0.0252
The proposed method	**93.10%**	**92.46%**	0.0150

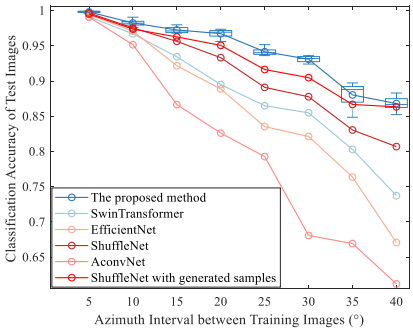


Fig. 18. Classification performances under various FSL conditions.

representation space under a higher visualization dimension. In conclusion, the CAE has learnt the intrinsic, diverse, and random attributes of images, as well as established effective associations among the classification hyperplanes of different classes, demonstrating its superior representation ability.

C. Evaluation of the Generalization Performance

In addition to representation ability, quantitative experiments are also carried out under various FSL conditions with the standard MSTAR dataset to evaluate the generalization performance.

1) Classification Performance Under FSL Conditions: First, the CAE is trained with 121 training samples. Then, the trained $\mathcal{E}$ is used to encode the test images to obtain predicted labels according to positions. Specially, classification experiments with AConvNet [3], ShuffleNet [42], EfficientNet [43], and SwinTransformer [44] are also carried out for comparisons. Classification accuracy and time complexity of different algorithms are listed in Table II. The proposed algorithm trained with only 12 samples per class achieves 93.1% and 92.46% classification accuracies for test data of 17° and 15° depression angles, far superior to the other algorithms but with comparable efficiency.

Furthermore, comparative experiments under various FSL conditions with angle intervals between training samples increasing in the range of [0°, 40°] have also been conducted. The results of these experiments are depicted in Fig. 18. The boxplot is adopted to visualize the statistics of subexperiments of the proposed method. When the angle interval between training samples is 5°, the proposed algorithm, CAE, achieves a comparable performance compared with other algorithms such as AConvNet, ShuffleNet, EfficientNet, and SwinTransformer, with a classification accuracy of 99.81%, 99.07%, 99.66%, 99.42%, and 99.12%, respectively. As the interval increases, the classification accuracy of test data gradually decreases, but the proposed algorithm still demonstrates a stable and superior performance among all the algorithms. Notably, the superiority of CAE becomes more apparent as the number of training samples decreases. When the angle interval is 40°, the proposed algorithm attains the best performance, outperforming the second-best algorithm, ShuffleNet, by 6.11%. In addition, the classification performances of ShuffleNet before and after data augmentation with generated samples were compared to evaluate the impact of samples generated by CAE on the performance of recognition model, whose results are presented in Fig. 18. It reveals that the generated samples can averagely boost the test accuracy by 5.65% compared with the model trained with pure training samples. Nevertheless, it still performs worse than CAE.

2) Angle Prediction Performance Under FSL Conditions: In addition, classification experiments are conducted to evaluate the angle generalization ability of the proposed CAE algorithm in comparison to other algorithms. The experiments are conducted with different angle intervals between the training and test samples, ranging from 1° to 30°. As shown in Fig. 19(a), the classification accuracy rate exhibits a pattern of initial decrease followed by slight fluctuations, and increase as the angular interval increased. The accuracy is observed to be the lowest when the azimuth angle interval is 15°. This trend can be attributed to the interval between training samples being close to 30°. The comparisons of the proposed CAE algorithm with other algorithms reveal that it has a more stable and superior classification performance, indicating a better generalization ability of azimuth angle. Furthermore, the variation in the angle prediction error with respect to the angle interval is shown in Fig. 19(b). The results of the experiments show that the azimuth angle prediction error varies between 13° and 33°. The t-SNE visualization of the distributions of the predicted and ground-truth angles is presented in Fig. 19(c), which demonstrates the favorable prediction performance of the proposed algorithm.

D. Ablation and Sensitivity Analysis of Noninterest Information

The introduction of the random terms z and η serves to embody the notions of diversity and randomness in the representation of images. To assess the efficacy of these terms, ablation and sensitivity analysis experiments were conducted under a variety of conditions. This included variations in the dimensions of z and the variance of η, with all the models trained using the same training samples and hyperparameter settings. As depicted in Fig. 20(a), the results indicate that the dimension of z primarily affects the variations in object appearance and imaging background. A satisfying generation effect was achieved with a dimension of 64, and as the

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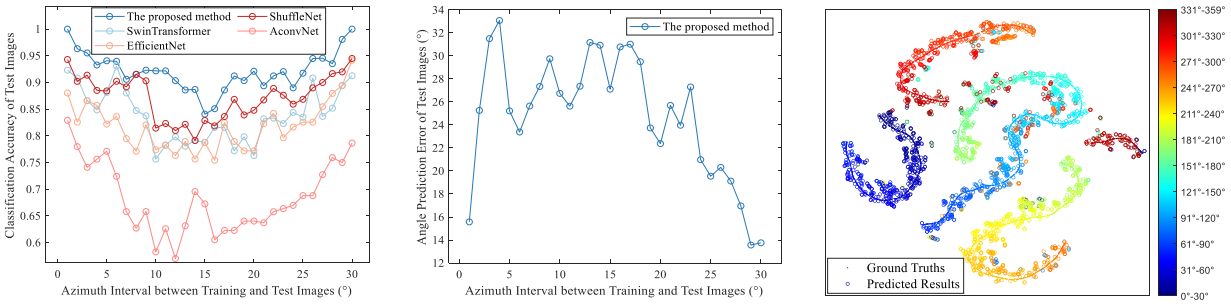


Fig. 19. Angle prediction results under FSL conditions. (a) Classification performance between different angle intervals. (b) Angle prediction performance with only 12 samples per class. (c) Visualization of predicted azimuth angles.

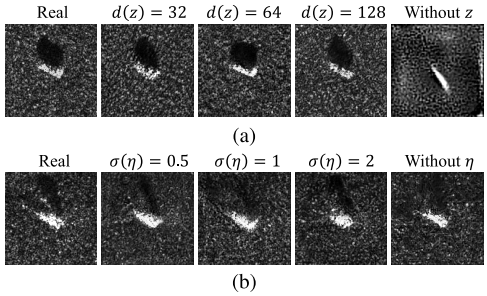


Fig. 20. Illustrations of the effect of random terms. (a) Effect of random term z. (b) Effect of random term η.

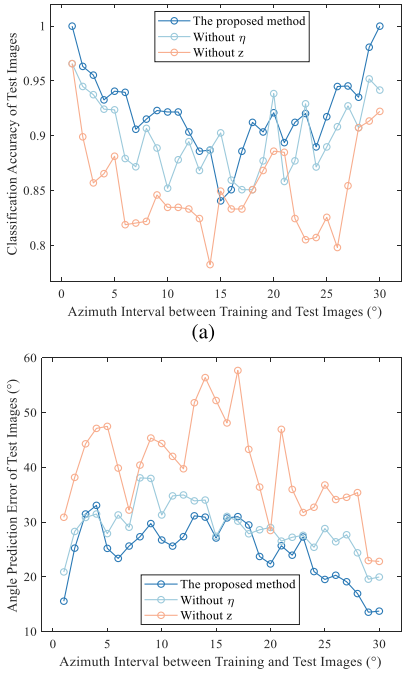


Fig. 21. Quantitative evaluation of the effect of random terms. (a) Effect of random terms on classification performance. (b) Effect of random terms on angle prediction performance.

dimension of z increases, the undulations of imaging scenes become more pronounced. Notably, the CAE was unable to generate vivid images without the presence of z, indicating that there is entangled information beyond the label and angle that needs to be represented by z. With regard to η, as presented in Fig. 20(b), the variance determines the randomness of background texture. As the variance increases from 0.5 to 2, the fluctuation of the background texture increases and its distribution becomes coarser. The experimental results indicate that the best results were achieved when the variance was set to 1, with the closest match to real images in terms of background texture. In the absence of η, the generated images were able to maintain the general contour and structure of the target image, but with distorted edges and coarser background. This suggests that the random information of images can be enhanced through the use of η, thereby improving the overall image fidelity.

Quantitative ablation experiments are further conducted to evaluate the effectiveness of z and η, as shown in Fig. 21. Compared with the networks without z and η, the CAE has achieved 4.49%, 3.08% classification performance improvement, and 9.06°, 4.22° angle prediction error reduction, respectively.

V. Conclusion

A causal perspective is taken on the representation of SAR image with neural networks, and a causal adversarial autoencoder is proposed for disentangled SAR target image representation and target recognition with sparse training samples. Different from the traditional adversarial generation models, it establishes an end-to-end optimization framework integrating generation, classification, and angle prediction in a low-dimensional space. The proposed CAE achieves manifold learning and probability distribution transformation through synergy of four subnetworks, i.e., mapper $\mathcal{M}$, generator $\mathcal{G}$, encoder $\mathcal{E}$, and discriminator $\mathcal{D}$. Disentangled distribution priors are introduced into the CAE following their casual relationships. Specifically, the CHE method is used to improve the sensitivity of angle representation. Besides, the SCE module is established to constrain the low-dimensional representations with consistent semantic properties. Experiments are carried out on the MSTAR and FUSAR-Ship datasets to evaluate the representation and generation performance of the proposed model. Quantitative comparisons demonstrate the effectiveness and superiority of the proposed method, which achieves classification accuracies of 93.10% and 92.46% for test data with 17° and 15° depression angles using only 12 training samples. The proposed CAE constructs a causally reasonable model, which shows superior performance with sparse training samples. On one hand, it can encode high-dimensional images into a low-dimensional space to obtain specific intrinsic attributes

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directly; on the other hand, it can generate a large number of high-quality samples conditioned on a set of disentangled semantic factors.

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